17	A Hubble telescope is used to observe two distant stars. The aperture diameter of the Hubble telescope is 2.4 m and visible light is used in the telescope. The two stars are observed to be just resolved. Which is the best estimate for the angular separation of the two stars?							
	A	1×10^{-8} rad	**B**	2×10^{-7} rad	**C**	1×10^{-5} rad	**D**	2×10^{-5} rad
L2	**Answer: B** **Rayleigh's criterion:** $\theta_{min} \approx \lambda / b$ where b = aperture diameter of the telescope = 2.4 m λ of visible light $\approx 500 \times 10^{-9}$ m ***Candidates expected to estimate average wavelength of visible light. Using 400 nm or 700 nm would also give final answer in the same order of magnitude.*** $\theta_{min} \approx \lambda / b = 500 \times 10^{-9} / 2.4 \approx 2 \times 10^{-7}$ **rad** **Since the two stars are just resolve,** **angular separation,** $\theta = \theta_{min} \approx 2 \times 10^{-7}$ **rad**							